

숙제3결과=>40점 중에 38점 획득

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- ① A rigid spacecraft rotates with a constant angular velocity whose components are given by $(0.1, -0.05, 0.03)$ rad/s when expressed in body fixed coordinate frame. Initially, the body fixed frame of spacecraft is exactly aligned with inertial frame.
- ① After 10s, what is the attitude of spacecraft in terms of rotation matrix, exponential coordinate, Euler angle, quaternion?
- ② After 10s, what is the angular velocity of spacecraft in terms of body fixed frame, inertial frame?

가분
 $A = \text{관성계 (IJK)}, B = \text{회전계 (ijk)}, W = W_{B/A} = W_{x1}\hat{i} + W_{y1}\hat{j} + W_{z1}\hat{k} = 0.1\hat{i} - 0.05\hat{j} + 0.03\hat{k} \text{ (rad/s)}$
 $\therefore \dot{W} = 0 = \text{각속도}$

$R = R_{B/A} \rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = R \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \rightarrow \text{관성계에 회전행렬을 곱하면 회전계 나옴.}$

①-i Rotation Matrix.
 $\dot{R} = S(W)R, R(t=0)_{3 \times 3} = R_0 \rightarrow \text{자세 운동식과 초기 자세 대다.}$
 $\text{자세 운동식을 풀면, } R(t=10) = e^{S(W)t} \cdot I_{3 \times 3} = \begin{pmatrix} 0.8452 & -0.4606 & -0.2616 \\ 0.0140 & 0.5132 & -0.8552 \\ 0.5246 & 0.7242 & 0.4417 \end{pmatrix}$

①-ii exponential coordinate.
 $\text{대각값} = 0.8452 + 0.5132 + 0.4417$
 $\phi = \cos^{-1} \left(\frac{\text{trace}(R) - 1}{2} \right) = 1.1576 \text{ (rad)} = 1.1576 \times 57.2958 \text{ (deg)} = 66.3247 \text{ (deg)}$
 $\alpha = \frac{1}{2 \sin \phi} \begin{pmatrix} R_{22} - R_{33} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{pmatrix} = \begin{pmatrix} 0.8639 \\ -0.4319 \\ 0.2592 \end{pmatrix} \rightarrow \text{exponential coordinate} \Rightarrow \alpha \phi = \begin{pmatrix} 1 \\ -0.5 \\ 0.3 \end{pmatrix}$

①-iii Euler angle.
 3-2-1 회전 순서로 보면, $R = R_{E/A} = R_1(\phi) R_2(\theta) R_3(\psi) = R(t=10)$

$$= \begin{pmatrix} \cos \psi \cos \theta \\ \cos \psi \sin \theta \sin \phi - \cos \psi \sin \theta \cos \phi \\ \sin \psi \sin \theta \sin \phi + \cos \psi \sin \theta \cos \phi \\ \sin \psi \cos \theta \\ \cos \psi \cos \theta \sin \phi - \sin \psi \cos \theta \cos \phi \\ \cos \psi \cos \theta \sin \phi + \sin \psi \cos \theta \cos \phi \\ \sin \psi \sin \theta \\ \sin \psi \cos \theta \sin \phi - \cos \psi \cos \theta \cos \phi \\ \cos \psi \cos \theta \sin \phi + \sin \psi \cos \theta \cos \phi \end{pmatrix} = \begin{pmatrix} 0.8452 & -0.4606 & -0.2616 \\ 0.0140 & 0.5132 & -0.8552 \\ 0.5246 & 0.7242 & 0.4417 \end{pmatrix}$$

 $\sin \theta = 0.2616 \rightarrow \theta = 15.1650^\circ$
 $\cos \theta \cdot \sin \phi = -0.4604 = 0.951 \cdot \sin \phi \rightarrow \sin \phi = \frac{-0.4604}{0.951} = -0.4840 \rightarrow \phi = -28.4846^\circ$
 $\frac{\cos \theta \cdot \sin \phi}{\cos \theta \cdot \cos \phi} = \tan \phi = \frac{-0.8552}{0.4417} = -1.9429 \rightarrow \phi = -62.7653^\circ$

①-iiii quaternion.
 $q_1 = a_1 \sin \frac{\phi}{2} = 0.3639, q_2 = a_2 \sin \frac{\phi}{2} = -0.4319, q_3 = a_3 \sin \frac{\phi}{2} = 0.2592, q_4 = \cos \frac{\phi}{2} = 0.8311$
 $\begin{pmatrix} 0.3639 \\ -0.4319 \\ 0.2592 \\ 0.8311 \end{pmatrix}$

②-: 회전계 (ijk)는 $W = W_{B/A} = 0.1\hat{i} - 0.05\hat{j} + 0.03\hat{k} \text{ (rad/s)} = \text{상수}$
 $W_{B/A}^B(t=10) = W_{B/A}^B(t=0) = (0.1, -0.05, 0.03) \text{ rad/s}$

②-: 관성계에서는 $W_{B/A}^A(t=10) = R_{A/B} \cdot W_{B/A}^B(t=10) = \begin{pmatrix} 0.8452 & -0.4606 & -0.2616 \\ 0.0140 & 0.5132 & -0.8552 \\ 0.5246 & 0.7242 & 0.4417 \end{pmatrix} \cdot \begin{pmatrix} 0.1 \\ -0.05 \\ 0.03 \end{pmatrix} = \begin{pmatrix} 0.1 \\ -0.05 \\ 0.03 \end{pmatrix} \text{ IJK frame qct.}$

② A rigid spacecraft has principal moments of inertia $(2000, 2000, 1000) \text{ kg-m}^2$ with no external torque acting on spacecraft. (2)

The third principal axis is axis of mass symmetry of spacecraft.

(a) What are all equilibrium of Euler eq?

(b) What is the time response for spacecraft angular velocity vectors, expressed in spacecraft body fixed $\mathcal{F}$, corresponding to initial condition $(0, 0, 0.8) \text{ (rad/s)}$ expressed in body fixed $\mathcal{F}$?

What is the time response for spacecraft attitude assuming that, initially, spacecraft body fixed $\mathcal{F}$ is aligned with inertial $\mathcal{I}$?

(c) Repeat part (b) for initial condition $(0.05, -0.05, 0) \text{ (rad/s)}$ in body fixed $\mathcal{F}$.

(Provide graphs of these time responses over whatever time interval you think demonstrates interesting response property (e.g. periodicity). You may use any suitable choice of coordinates to describe the spacecraft attitude responses.)

(a) $(I_x, I_y, I_z) = (2000, 2000, 1000) \text{ kg-m}^2$ $(I_x, I_y, I_z) = (0, 0, 0)$ 2축은 동등 대칭축이다. $I = I_x = I_y \neq I_z$.
axis-symmetric

$$\begin{aligned} I_x \dot{\omega}_x &= (I_y - I_z) \omega_y \omega_z \\ I_y \dot{\omega}_y &= (I_z - I_x) \omega_z \omega_x \\ I_z \dot{\omega}_z &= 0, \quad \dot{\omega}_z = \text{상수} \end{aligned} \rightarrow \text{평면 회전하면 위의 두 식도 0이 되는데 I는 상수라서 } \omega_y \omega_z = 0 \text{ 이거나 한다.}$$

$$\omega_x \omega_z = 0$$

$$\therefore \omega_z = \text{상수} = 0 \text{ 이면 된다.}$$

or

$$\omega_z \neq 0, \text{ then } \omega_x = \omega_y = 0$$

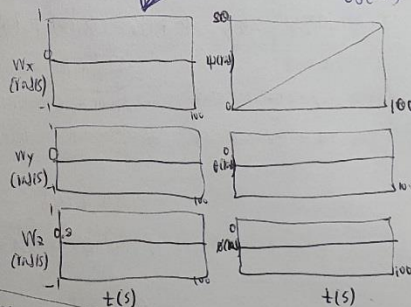
$$\therefore \omega = \begin{pmatrix} \omega_x \\ \omega_y \\ 0 \end{pmatrix}^c, \quad \omega = \begin{pmatrix} 0 \\ 0 \\ \omega_z \end{pmatrix}^b \text{ 이다.}$$

(b) $\omega(t=0) = \begin{pmatrix} 0 \\ 0 \\ 0.8 \end{pmatrix}$ 은 equilibrium solution이다. 그래서 각속도는 상수다. $\omega(t) = \omega(t=0) = \begin{pmatrix} 0 \\ 0 \\ 0.8 \end{pmatrix} \Rightarrow$ 주기 각속도 유지.

$$\begin{aligned} \dot{\psi} &= (\omega_y \sin \theta + \omega_z \cos \theta) \sec \theta = \omega_z \cos \theta \sec \theta \\ \dot{\theta} &= \omega_y \cos \theta - \omega_z \sin \theta = -\omega_z \sin \theta \\ \dot{\phi} &= \omega_x + \omega_y \sin \theta \tan \theta + \omega_z \cos \theta \tan \theta = \omega_z \cos \theta \tan \theta \end{aligned}$$

$$\begin{aligned} \psi(0) &= 0 \\ \theta(0) &= 0 \\ \phi(0) &= 0 \end{aligned}$$

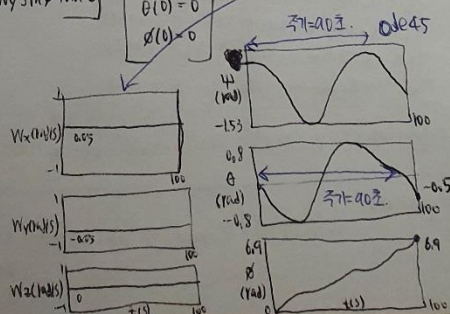
$$\begin{aligned} \dot{\psi}(t) &= \omega_z = \text{상수} \\ \dot{\theta}(t) &= 0 = \text{상수} \\ \dot{\phi}(t) &= 0 = \text{상수} \end{aligned} \rightarrow \begin{aligned} \psi(t) &= \omega_z t = 0.8t \\ \theta(t) &= 0 \\ \phi(t) &= 0 \end{aligned} \quad \left(\because \text{주기 } \theta(0)=0 \text{ 이라서 } \right)$$



(c) $\omega(t=0) = \begin{pmatrix} 0.05 \\ -0.05 \\ 0 \end{pmatrix}$ 은 equilibrium solution이다. 그래서 각속도는 상수다. $\omega(t) = \omega(t=0) = \begin{pmatrix} 0.05 \\ -0.05 \\ 0 \end{pmatrix} \Rightarrow$ 주기 각속도 유지.

$$\begin{aligned} \dot{\psi} &= (\omega_y \sin \theta + \omega_z \cos \theta) \sec \theta = \omega_y \sin \theta \sec \theta \\ \dot{\theta} &= \omega_y \cos \theta - \omega_z \sin \theta = \omega_y \cos \theta \\ \dot{\phi} &= \omega_x + \omega_y \sin \theta \tan \theta + \omega_z \cos \theta \tan \theta = \omega_x + \omega_y \sin \theta \tan \theta \end{aligned}$$

$$\begin{aligned} \psi(0) &= 0 \\ \theta(0) &= 0 \\ \phi(0) &= 0 \end{aligned}$$



③ A rigid spacecraft has principal moments of inertia $(2000, 1500, 1000) \text{ kg-m}^2$ with no external torque acting on spacecraft. The axis is axis of symmetry of spacecraft.

① What are all equilibrium of Euler eq.?

② What is the time response for spacecraft angular velocity vector, expressed in spacecraft body fixed $\mathcal{B}$, corresponding to initial condition $(0, 0, 0.8) \text{ (rad/s)}$ expressed in body fixed $\mathcal{B}$?

③ Repeat part ② for initial condition $(0.05, -0.05, 0) \text{ (rad/s)}$ in body fixed $\mathcal{B}$.

(Provide graphs of these time responses over whatever time interval you think demonstrates interesting response property). You may use any suitable choice of coordinates to describe the spacecraft attitude responses.

④ $(I_x, I_y, I_z) = (2000, 1500, 1000) \text{ kg-m}^2$ $(M_x, M_y, M_z) = (0, 0, 0)$ 비대칭이다. $I_x \neq I_y \neq I_z$.
symmetric

$$I_x \dot{w}_x = (I_y - I_z) w_y w_z$$

$$I_y \dot{w}_y = (I_z - I_x) w_z w_x$$

$$I_z \dot{w}_z = (I_x - I_y) w_x w_y$$

→ 편미하면 3개 식이 나오는데 개개

$$w_y w_z = 0$$

$$w_z w_x = 0$$

$$w_x w_y = 0$$

$$w = \begin{pmatrix} 0 \\ 0 \\ 0.8 \end{pmatrix}$$

$$w = \begin{pmatrix} 0 \\ w_y \\ 0 \end{pmatrix}$$

$$w = \begin{pmatrix} w_x \\ 0 \\ 0 \end{pmatrix}$$

이 경우들이 있다.

⑤ $w(t=0) = \begin{pmatrix} 0 \\ 0 \\ 0.8 \end{pmatrix}$ 은 equilibrium solution 이다. 그래서 각속도는 상수다. $w(t) = w(t=0) = \begin{pmatrix} 0 \\ 0 \\ 0.8 \end{pmatrix} \Rightarrow$ 초기 각속도 유지.

오일러 각 자세 벡터로,

$$\begin{aligned} \psi &= (w_y \sin \theta + w_z \cos \theta) \sec \phi \\ \theta &= w_y \cos \phi - w_z \sin \phi \\ \phi &= w_x + w_y \sin \phi \tan \theta + w_z \cos \phi \tan \theta \end{aligned}$$

$$\begin{aligned} \psi &= w_z \cos \theta \sec \phi \\ \theta &= -w_z \sin \theta \\ \phi &= w_z \cos \theta \tan \theta \end{aligned}$$

$$\begin{aligned} \psi(0) &= 0 \\ \theta(0) &= 0 \\ \phi(0) &= 0 \end{aligned}$$

$$\dot{\psi}(t) = w_z = \text{상수}$$

$$\dot{\theta}(t) = 0 = \text{상수}$$

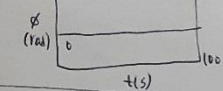
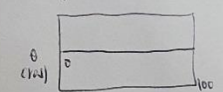
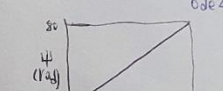
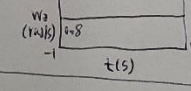
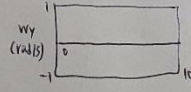
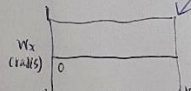
$$\dot{\phi}(t) = 0 = \text{상수}$$

$$\psi(t) = w_z t = 0.8t$$

$$\theta(t) = 0$$

$$\phi(t) = 0$$

(-3개씩 $\theta=0$ $\phi=0$ $\psi=0.8t$)



⑥ $w(t=0) = \begin{pmatrix} 0.05 \\ -0.05 \\ 0 \end{pmatrix}$ 은 NOT equilibrium solution 이다. 따라서 아래 식들을 수치해석적으로 propagation 한다.

$$\dot{w}_x = \frac{I_y - I_z}{I_x} w_y w_z$$

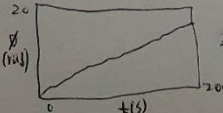
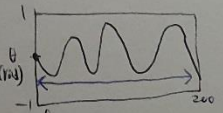
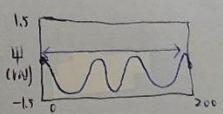
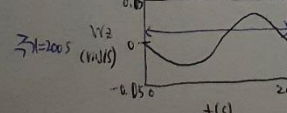
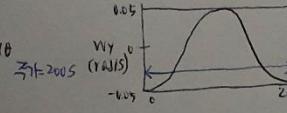
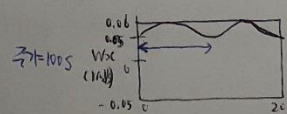
$$\dot{w}_y = \frac{I_z - I_x}{I_y} w_z w_x$$

$$\dot{w}_z = \frac{I_x - I_y}{I_z} w_x w_y$$

$$\dot{\psi} = (w_y \sin \theta + w_z \cos \theta) \sec \phi$$

$$\dot{\theta} = w_y \cos \phi - w_z \sin \phi$$

$$\dot{\phi} = w_x + w_y \sin \phi \tan \theta + w_z \cos \phi \tan \theta$$



④ optional credit point. Develop computer animations of the spacecraft motions in the above problem 5, so that the spacecraft velocity vector and the spacecraft attitude can be visualized. You may use Matlab.

등속회살표는 관성계, 궤도회살표는 회전계, 행광회살표는 오메가이다. 아래 모든 그림에 동일하게 적용된다. 그림순서는 2-b 2-c 이다. 3-b 3-c 이다.

